Analysis 2
Serie 10
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Exercise 1

$$X(\varphi,\psi) = (\cos\varphi\cos\psi, \cos\varphi\sin\psi, \sin\varphi), \quad -\frac{\pi}{2} < \varphi_1 < \varphi_2 < \frac{\pi}{2}, -\pi < \psi_1 < \psi_2 < \pi$$

$$J_X = \begin{pmatrix} -\sin\varphi\cos\psi & -\cos\varphi\sin\psi \\ -\sin\varphi\sin\psi & \cos\varphi\cos\psi \\ \cos\varphi & 0 \end{pmatrix}$$

$$\sqrt{\det(J_X^T J_X)} = \sqrt{\det\left(\begin{pmatrix} 1 & 0 \\ 0 & \cos^2\varphi \end{pmatrix}\right)} = \cos\varphi$$

$$\Rightarrow \partial X = \int_{\varphi_1}^{\varphi_2}\int_{\psi_1}^{\psi_2} \cos\varphi d\psi d\varphi = (\psi_2 - \psi_1)(\sin\varphi_2 - \sin\varphi_1)$$

Exercise 2

$$-1 = |(q,w)| = q^2 + w^2 = 1, \quad x^2 + y^2 = 4$$

$$\Rightarrow x = r \quad \sin(\varphi) \quad y = r\cos(\varphi) \quad q = \sin(\alpha) \quad w = \cos(\alpha)$$

$$\Rightarrow \partial T = \int_0^{2\pi}\int_0^{2\pi} 2 d\varphi d\alpha = 8\pi^2$$

Exercise 3

1. Ω open, $\partial_{\text{side}}\Omega$ closed $\Rightarrow$ not connected
2.

$$\text{vol}_3(\Omega) = \int_0^{2\pi}\int_a^b\int_0^{f(x)} r dr dx d\varphi = 2\pi\int_a^b \frac{1}{2} f(x)^2 dx = \int_a^b f(x)^2 dx$$

3.

$$J^T = \begin{pmatrix} 1 & f'(x)\cos\theta & f'(x)\sin\theta \\ 0 & -f(x)\sin\theta & f(x)\cos\theta \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 0 \\ f'(x)\cos\theta & -f(x)\sin\theta \\ f'(x)\sin\theta & f(x)\cos\theta \end{pmatrix}$$

$$\Rightarrow \det(J^T J) = \det\left(\begin{pmatrix} 1 & +f'(x)^2 & 0 \\ 0 & f(x)^2 \end{pmatrix}\right) = (1 + f'(x)^2) f(x)^2$$

$$\Rightarrow \int_0^{2\pi}\int_a^b \sqrt{1 + f'(x)^2} f(x) dx = 2\pi\int_a^b \sqrt{1 + f'(x)^2} f(x) dx$$

4.

$$f(x) = \frac{1}{x}, \quad a = 1, \quad b = \infty$$

$$\Rightarrow A = 2\pi\int_1^\infty \sqrt{1 + \frac{1}{x^4}}\frac{1}{x} dx = 2\pi\int_1^\infty \sqrt{\frac{x^4 + 1 + x^2}{x^4}} dx$$

$$\leq 2\pi\int_0^\infty \frac{1}{x} dx = \infty$$

$$V = \pi\int_1^\infty \frac{1}{x^2} dx = \pi\left[-\frac{1}{x}\right]_1^\infty = \pi$$

Exercise 4

1.

$$\frac{d}{dt}(\sin(t)) = \cos(t)$$

$$\begin{aligned}
\frac{d}{dt}\left(\cos(t) + \log\left(\tan\left(\frac{t}{2}\right)\right)\right) &= -\sin(t) + \frac{1}{\tan(\frac{t}{2})} \cdot \frac{1}{2} \cdot \frac{1}{\cos(\frac{t}{2})^2} \\
&= -\sin(t) + \frac{1}{2}\frac{1}{\sin(\frac{t}{2})\cos(\frac{t}{2})} \\
&= \frac{1}{\sin(t)} - \sin(t) = \frac{\cos(t)}{\tan(t)}
\end{aligned}$$

$$\begin{aligned}
\Rightarrow L\left(\sigma \mid_{[\varepsilon, \frac{\pi}{2}]}\right) &= \int_{\varepsilon}^{\frac{\pi}{2}} \sqrt{\cos^2(t) + \cos^2 \frac{1}{\tan^2(t)}} dt \\
&= \int_{\varepsilon}^{\frac{\pi}{2}} \sqrt{\frac{1}{\tan^2(t)}}) dt \\
&= \int_{\varepsilon}^{\frac{\pi}{2}} \cot(t) dt \\
&= \log(\sin(t))|_{\varepsilon}^{\frac{\pi}{2}}
\end{aligned}$$

$$\Rightarrow \lim_{\varepsilon \to 0^+} \log(\sin(\varepsilon)) = \lim_{\varepsilon \to 0^+} \log(\varepsilon) = \infty$$

$$\Rightarrow L \to \infty \quad \text{as } \varepsilon \to 0$$

2.

from 1.

$$\sigma(t) = \left(\sin(t), \log\left(\tan\left(\frac{t}{2}\right)\right)\right)$$

$$\sigma'(t) = \left(\cos(t), \frac{\cos^2(t)}{\sin(t)}\right)$$

$$\Rightarrow t(t)(x) = \sigma(t) + x\sigma'(t) = \left(x\cos(t) + \sin(t), \log\left(\tan\left(\frac{t}{2}\right)\right) + x\frac{\cos(t)}{\tan(t)}\right)$$

Intersection with y-axis

$$\sin(t) = -x\cos(t) \Rightarrow x = -\tan(t)$$

$$\Rightarrow I_t = \int_{-\tan(t)}^{0} \sqrt{\cos^2(t) + \frac{\cos^2(t)}{\tan^2(t)}} dx = \frac{\tan(t)}{\tan(t)} = 1$$

Exercise 5

$$\gamma(t) = \left(ae^{bt}\cos(t), ae^{bt}\sin(t)\right)$$

$$\gamma'(t) = \left(ae^{bt}(b - \sin(t)), ae^{bt}(b + \cos(t))\right)$$

$$
\begin{aligned}
L\left(\gamma \mid_{[0,T]}\right) &= \int_0^T ae^{bt}\sqrt{(b-\sin(t))^2+(b+\cos(t))^2}dt \\
&= \int_0^T ae^{bt}\sqrt{b^2+1}dt = a\sqrt{b^2+1}\frac{1}{b}e^{bt}|_0^T \\
&\Rightarrow \lim_{T\to\infty}\frac{a}{b}\sqrt{b^2+1}e^{bT}|_0^T = -\frac{a}{b}\sqrt{b^2+1}
\end{aligned}
$$

2.

Exercise 6

1.

$$
J_\Phi = \begin{pmatrix} a\cos(t) & -as\sin(t) \\ b\sin(t) & bs\cos(t) \end{pmatrix} \quad J_\Phi^T = \begin{pmatrix} a\cos(t) & b\sin(t) \\ -as\sin(t) & bs\cos(t) \end{pmatrix}
$$

$$
\begin{aligned}
\sqrt{J_\Phi J_\Phi^T} &= \sqrt{(a^2s^2\sin^2 + a^2\cos^2)(b^2s^2\cos^2 + b^2\sin^2) - (ab\sin\cos - abs^2\cos\sin)(ab\cos\sin - abs^2\sin\cos)} \\
&= \sqrt{abs\cos^2(t) - (-abs\sin^2(t))} = abs
\end{aligned}
$$

2.

$$
x = as\cos(t) \quad y = bs\sin(t)
$$

$$
x^2+y^2 = a^2s^2\cos^2(t) + b^2s^2\sin^2(t)
$$

$$
\begin{aligned}
J_0 &= \int_0^{2\pi}\int_0^1 (a^2s^2\cos^2(t) + b^2s^2\sin^2(t))absdsdt \\
&= ab\int_0^1 s^3\left(\int_0^{2\pi}\cos^2(t)dt + \int_0^{2\pi}\sin^2(t)dt\right)ds \\
&= \frac{ab\pi}{4}(a^2+b^2)
\end{aligned}
$$